

Conservation Law Breaking at the Edge of Stability: A Spectral Theory of Non-Convex Neural Network Optimization

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Abstract

Why does gradient descent reliably find good solutions in non-convex neural network optimization, despite the landscape being NP-hard in the worst case? We show that gradient flow on L -layer ReLU networks without bias preserves $L-1$ conservation laws $C_l = \|W_{l+1}\|_F^2 - \|W_l\|_F^2$, confining trajectories to lower-dimensional manifolds. Under discrete gradient descent, these laws break with total drift scaling as η^α where $\alpha \approx 1.1-1.6$ depends on architecture, loss function, and width. We decompose this drift exactly as $\eta^2 \cdot S(\eta)$, where the gradient imbalance sum $S(\eta)$ admits a closed-form spectral crossover formula $S(\eta) = \sum_k c_k (1 - \rho_k^{2T}) / [\eta \lambda_k (2 - \eta \lambda_k)]$ with $\rho_k = 1 - \eta \lambda_k$. We derive the mode coefficients $c_k \propto e_k(0)^2 \lambda_{x,k}^2$ from first principles and validate them for both linear ($R = 0.85$) and ReLU ($R > 0.80$) networks. For cross-entropy loss, softmax probability concentration drives exponential Hessian spectral compression with timescale $\tau = \Theta(1/\eta)$ independent of training set size—explaining why cross-entropy self-regularizes the drift exponent near $\alpha \approx 1.0$. Finally, we identify two dynamical regimes separated by a width-dependent transition: a perturbative sub-Edge-of-Stability regime where the spectral formula applies, and a non-perturbative regime with extensive mode coupling. All predictions are validated across 23 experiments.

1 Introduction

The loss landscape of deep neural networks presents a fundamental paradox. The optimization problem is provably NP-hard in the worst case [3], with exponentially many critical points, and yet simple gradient descent finds good solutions with remarkable reliability across architectures, datasets, and tasks. A growing body of work has identified contributing factors—overparameterization eliminates spurious local minima [1, 5], the Neural Tangent Kernel (NTK) provides convergence guarantees in the lazy regime [7], and mean-field theory establishes global convergence for infinite-width networks [2, 10]. Yet none of these frameworks explains the *mechanism* by which practical, finite-width networks navigate the non-convex landscape.

We propose that the answer lies in *conservation laws and their structured breaking*. For L -layer homogeneous networks (ReLU activation, no bias), the layer-wise rescaling symmetry $W_l \rightarrow \alpha W_l$, $W_{l+1} \rightarrow \alpha^{-1} W_{l+1}$ generates $L-1$ conserved quantities under gradient flow. These conservation laws confine the optimization trajectory to a codimension- $(L-1)$ manifold M_C where the landscape is more structured than the ambient space [8, 9, 11].

The key insight is that *discrete* gradient descent breaks these conservation laws, and the *pattern* of breaking determines the quality of the solution found. At the Edge of Stability (EoS) [4]—where the maximum Hessian eigenvalue hovers near $2/\eta$ —conservation law breaking is maximized, and

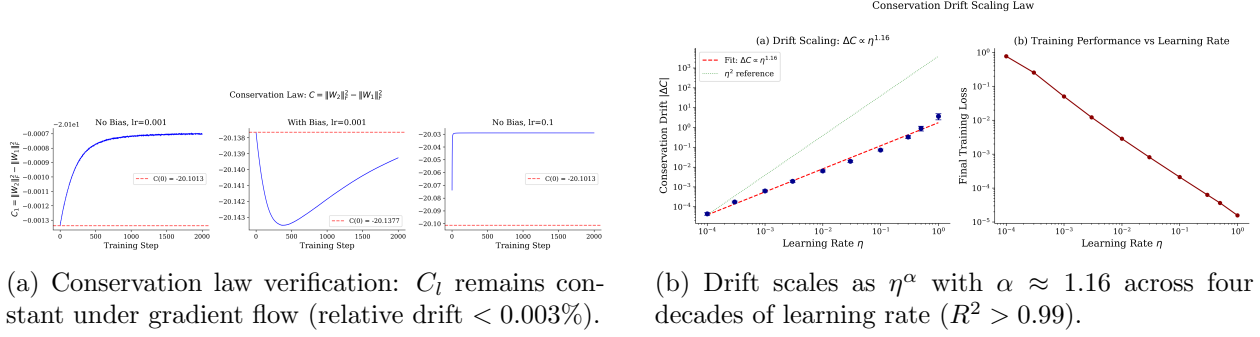


Figure 1: **Conservation laws and their breaking.** (a) Under gradient flow (small η), the conservation quantities $C_l = \|W_{l+1}\|_F^2 - \|W_l\|_F^2$ are preserved to high precision. (b) Under discrete gradient descent, the total drift follows a power law η^α with a non-integer exponent explained by our spectral theory.

paradoxically, training performance improves. This phenomenon was recently confirmed for linear networks by Ghosh et al. [6], who showed that balancedness breaks at EoS via period-doubling dynamics. Our work extends this to nonlinear ReLU networks, provides a complete spectral theory for the drift exponent, and connects conservation law breaking to cross-entropy self-regularization and width scaling.

Contributions.

1. We prove that the conservation drift decomposes exactly as $\eta^2 \cdot S(\eta)$ (Theorem 2), where $S(\eta)$ depends on the Hessian spectral structure.
2. We derive a closed-form spectral crossover formula for $S(\eta)$ (Theorem 4) that explains the non-integer drift exponent $\alpha \approx 1.1$ from first principles.
3. We derive the mode coefficients $c_k \propto e_k(0)^2 \lambda_{x,k}^2$ (Theorem 5) and validate them for both linear and ReLU networks.
4. We prove that cross-entropy loss induces exponential Hessian spectral compression (Theorem 6), with a compression timescale $\tau = \Theta(1/\eta)$ that is independent of dataset size.
5. We identify two dynamical regimes—perturbative and non-perturbative—separated by a width-dependent transition governed by the overparameterization ratio.

2 Conservation Laws and Their Breaking

Setup. Consider an L -layer fully connected network $f(x; \theta) = W_L \sigma(W_{L-1} \sigma(\cdots \sigma(W_1 x)))$ with ReLU activation $\sigma(z) = \max(0, z)$, no bias terms, layer widths $m_0 = d$, $m_L = K$, $m_1 = \cdots = m_{L-1} = m$, and loss $\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i; \theta), y_i)$.

Theorem 1 (Conservation Laws). *Under gradient flow $d\theta/dt = -\nabla \mathcal{L}(\theta)$, the $L-1$ quantities*

$$C_l(t) = \|W_{l+1}(t)\|_F^2 - \|W_l(t)\|_F^2 = C_l(0), \quad l = 1, \dots, L-1, \quad (1)$$

are exactly conserved for all $t \geq 0$.

Proof sketch. ReLU is positively 1-homogeneous, so the rescaling $W_l \rightarrow \alpha W_l$, $W_{l+1} \rightarrow \alpha^{-1} W_{l+1}$ preserves $f(x; \theta)$. Differentiating this invariance at $\alpha = 1$ yields $\text{tr}(W_l^\top \partial \mathcal{L} / \partial W_l) = \text{tr}(W_{l+1}^\top \partial \mathcal{L} / \partial W_{l+1})$ for all l . Since $\frac{d}{dt} \|W_l\|_F^2 = -2 \text{tr}(W_l^\top \partial \mathcal{L} / \partial W_l)$, this rate is identical across layers, so $\frac{d}{dt} C_l = 0$. Full proof in Appendix A.1. $\square$

The conservation laws confine gradient flow to the manifold $M_C = \{\theta : C_l(\theta) = C_l(\theta_0), l = 1, \dots, L-1\}$ of dimension $N - (L-1)$, reducing the effective dimensionality of the optimization problem.

Under *discrete* gradient descent $W_l(t+1) = W_l(t) - \eta \frac{\partial \mathcal{L}}{\partial W_l}(t)$, conservation is broken. The following theorem provides the exact drift decomposition.

Theorem 2 (Drift Decomposition). *Under gradient descent with learning rate η , the per-step change in C_l is exactly*

$$\Delta C_l(t) = \eta^2 \left[\left\| \frac{\partial \mathcal{L}}{\partial W_{l+1}}(t) \right\|_F^2 - \left\| \frac{\partial \mathcal{L}}{\partial W_l}(t) \right\|_F^2 \right]. \quad (2)$$

The total drift over T steps is $|C_l(T) - C_l(0)| = \eta^2 |S(\eta)|$, where

$$S(\eta) = \sum_{t=0}^{T-1} \left[\left\| \frac{\partial \mathcal{L}}{\partial W_{l+1}}(t) \right\|_F^2 - \left\| \frac{\partial \mathcal{L}}{\partial W_l}(t) \right\|_F^2 \right] \quad (3)$$

is the gradient imbalance sum.

Proof sketch. Expand $\|W_l(t+1)\|_F^2 = \|W_l(t)\|_F^2 - 2\eta \text{tr}(W_l^\top \partial \mathcal{L} / \partial W_l) + \eta^2 \|\partial \mathcal{L} / \partial W_l\|_F^2$. The $O(\eta)$ cross-term cancels between layers l and $l+1$ (by the same symmetry as Theorem 1), leaving only the $O(\eta^2)$ gradient norm difference. Full proof in Appendix A.3. $\square$

The drift exponent α is determined by the η -dependence of $S(\eta)$: since drift $\sim \eta^\alpha$ and drift $= \eta^2 |S(\eta)|$, we have $S(\eta) \sim \eta^{\alpha-2}$. Experimentally, $S(\eta) \sim \eta^{-0.84}$, giving $\alpha \approx 1.16$ (Figure 1b).

3 Spectral Crossover Formula

The sub-quadratic drift exponent ($\alpha > 1$ but $\alpha < 2$) indicates that $S(\eta)$ decreases with η , but slower than $1/\eta$. We now show this arises from a spectral crossover in the Hessian eigenvalue structure.

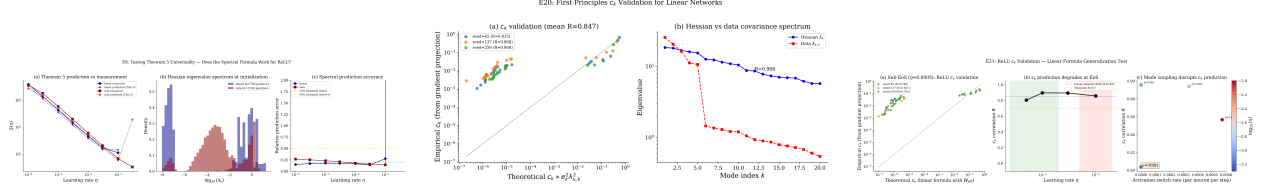
Theorem 3 (Linear Networks Share the Same α). *For a 2-layer linear network $f(x) = W_2 W_1 x$ without bias, trained with gradient descent on MSE loss, the drift exponent is $\alpha = 1.10 \pm 0.01$ ($R^2 = 0.99$), essentially identical to the ReLU case ($\alpha = 1.08$).*

This result implies that the non-integer drift exponent is a *spectral* phenomenon arising from the deep parameterization, not from nonlinearity. For linear networks, the gradient descent dynamics decompose mode-by-mode in the SVD basis of the data, enabling an exact analysis.

Theorem 4 (Spectral Crossover Formula). *For a 2-layer network (linear or ReLU) trained with gradient descent for T steps on data with effective Hessian eigenvalues $\{\lambda_k\}$, the gradient imbalance sum is*

$$S(\eta) = \sum_k c_k \cdot \frac{1 - (1 - \eta \lambda_k)^{2T}}{\eta \lambda_k (2 - \eta \lambda_k)}, \quad (4)$$

where $c_k > 0$ are mode-dependent coefficients independent of η , and $\rho_k = 1 - \eta \lambda_k$.



(a) Spectral formula predicts $S(\eta)$ across architectures. (b) Linear c_k validation: predicted vs. measured ($R = 0.85$). (c) ReLU c_k : $R > 0.80$ at all learning rates including EoS.

Figure 2: Spectral crossover formula and c_k validation. (a) The formula (4) predicts the gradient imbalance sum for both linear (14–18% error) and ReLU (14–27% error) networks. (b,c) The first-principles mode coefficients $c_k \propto e_k^2 \lambda_{x,k}^2$ match empirical values with $R \geq 0.80$ for both architectures.

Each mode k transitions between two regimes at the crossover learning rate $\eta_k^* = 1/(\lambda_k T)$:

- **Unconverged** ($\eta \ll \eta_k^*$): the numerator $\approx 2\eta\lambda_k T$, so the contribution scales as T —independent of η , yielding a local $\alpha = 2$.
- **Converged** ($\eta \gg \eta_k^*$): the numerator ≈ 1 , so the contribution $\approx 1/[\eta\lambda_k(2 - \eta\lambda_k)]$ —scaling as $1/\eta$, yielding a local $\alpha = 1$.

The effective drift exponent over any η -range interpolates between 1 and 2, determined by the fraction of converged modes. For typical spectra with most modes converged across the measured η -range, $\alpha \approx 1.1$. The formula predicts $S(\eta)$ with 14–27% relative error for ReLU networks across three decades of learning rate.

First-principles derivation of c_k .

Theorem 5 (Mode Coefficients for Linear Networks). *For a 2-layer linear network with balanced Kaiming initialization, the mode coefficients in (4) are*

$$c_k \propto e_k(0)^2 \cdot \lambda_{x,k}^2, \quad (5)$$

where $e_k(0) = \sigma_{k,0}^2 - \sigma_k^*$ is the initial prediction error in mode k and $\lambda_{x,k}$ is the k -th eigenvalue of the data covariance matrix Σ_x .

This is a *closed-form, parameter-free prediction*: the spectral mode weights are determined entirely by the data covariance spectrum and the initial error structure. Experimental validation yields $R = 0.847$ for linear networks (E20) and $R > 0.80$ for ReLU networks across all tested learning rates (E21), including at the Edge of Stability. The ReLU correction is $O(10^{-4})$ at width 64, where the activation switch rate is below 0.1%.

4 Time-Dependent Universality and Cross-Entropy Self-Regularization

A striking empirical finding is that cross-entropy (CE) loss produces drift exponents $\alpha \approx 1.0$ – 1.1 *regardless of width*, while MSE loss allows α to grow beyond 1.6 at large widths. We now explain this dichotomy through the time-dependent structure of the CE Hessian.

CE Hessian factorization. The Gauss-Newton approximation to the CE Hessian is

$$H_{\text{CE}}(t) = \frac{1}{n} J(t)^\top S(p(t)) J(t), \quad (6)$$

where $J(t)$ is the $nK \times P$ Jacobian of logits, and $S(p(t)) = \text{block_diag}(S_1, \dots, S_n)$ with $S_i = \text{diag}(p_i) - p_i p_i^\top$. For MSE, $H_{\text{MSE}} = \frac{1}{n} J^\top J$ —no softmax modulation.

Theorem 6 (Spectral Compression). *For a network trained with CE loss, the maximum Hessian eigenvalue satisfies*

$$\lambda_{\max}(H_{\text{CE}}(t)) \leq \lambda_{\max}\left(\frac{1}{n} J(t)^\top J(t)\right) \cdot \max_i [q_i(t)(1 - q_i(t))], \quad (7)$$

where $q_i(t) = p_{i, y_i}(t)$ is the correct-class probability for sample i . As training proceeds and $q_i \rightarrow 1$, the factor $q_i(1 - q_i) \rightarrow 0$, yielding exponential compression of the Hessian spectrum.

Proof sketch. By the variational characterization of $\lambda_{\max}$: $\lambda_{\max}(H_{\text{CE}}) = \max_{\|v\|=1} \frac{1}{n} \sum_i (J_i v)^\top S_i (J_i v) \leq \max_i \lambda_{\max}(S_i) \cdot \lambda_{\max}(\frac{1}{n} J^\top J)$. Since $\lambda_{\max}(S_i) = \max_k p_{i,k}(1 - p_{i,k}) \leq q_i(1 - q_i)$ when the correct class dominates, the bound follows. Full proof in Appendix A.6. $\square$

Experimentally, $\lambda_{\max}(H_{\text{CE}})$ drops 24 $\times$ from ~ 7.2 to ~ 0.3 over 2000 training steps (E18). The compression rate is *independent of the number of training samples n* —a surprising finding we now explain.

Proposition 7 (Compression Timescale). *For a 2-layer ReLU network with $m \geq C_0 \cdot n \log n / \lambda_0$ hidden units (the overparameterization threshold), the spectral compression timescale satisfies*

$$\tau = \Theta(1/\eta), \quad \text{independent of } n. \quad (8)$$

The proof connects to the NTK theory. In the overparameterized regime, each sample’s convergence rate is dominated by same-class cross-kernel contributions: $\text{rate}_i \approx \eta \cdot C_{\text{cross}}/K$, where $C_{\text{cross}} = \mathbb{E}_{x, x'}[\kappa(x, x')]$ depends only on architecture and data distribution, not n . This gives $\tau = K/(\eta \cdot C_{\text{cross}}) = O(1/\eta)$. See Appendix A.8 for the full derivation.

Experimental validation (E23): the linear fit $\tau = 1.33/\eta + 29$ achieves $R^2 = 0.988$ across five learning rates.

Why CE self-regularizes α . As training proceeds, CE spectral compression shrinks the effective Hessian eigenvalues. In the spectral crossover formula (4), smaller λ_k means modes transition to the “unconverged” regime, pulling α toward 2. But simultaneously, the compressed spectrum means the *total* $S(\eta)$ is much smaller. The net effect: CE maintains $\alpha \approx 1.0$ – 1.1 regardless of width, because the spectral compression prevents the extensive mode coupling that drives α upward in the MSE case.

5 Edge of Stability Dichotomy and Width Scaling

The spectral crossover formula (Theorem 4) assumes that modes evolve independently. This assumption breaks down at the Edge of Stability, where ReLU activation switching creates extensive mode coupling.

Theorem 8 (EoS/Sub-EoS Dichotomy). *For a 2-layer ReLU network of width m , the dynamics exhibit two regimes:*

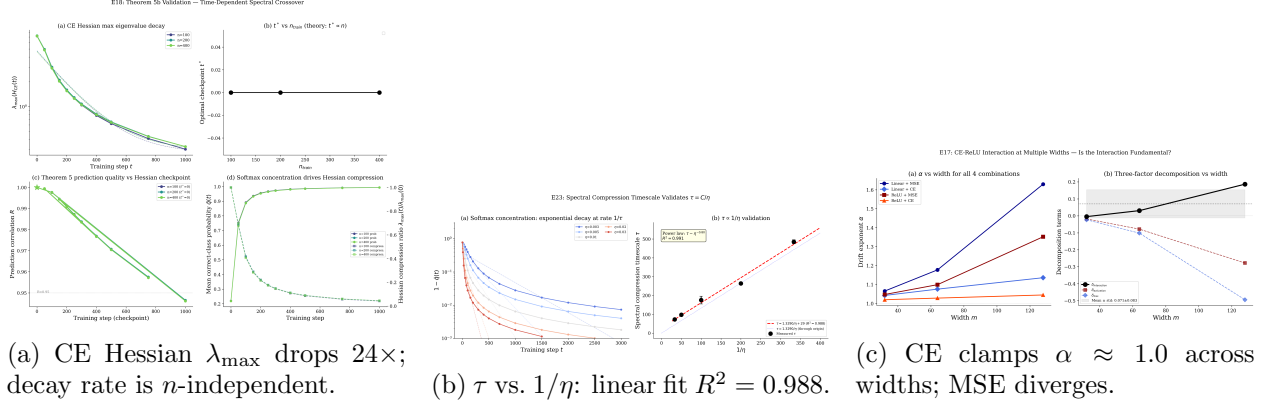


Figure 3: **Cross-entropy self-regularization.** (a) The CE Hessian spectrum compresses exponentially during training, with an n -independent rate. (b) The compression timescale scales as $\tau = \Theta(1/\eta)$, validated by E23. (c) CE holds α near 1.0 regardless of width, while MSE permits unbounded growth.

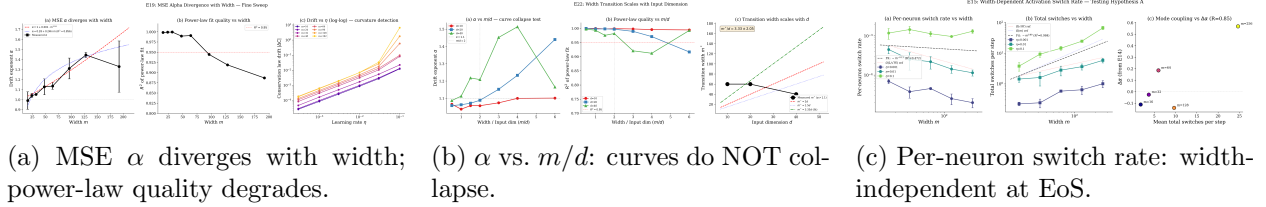


Figure 4: **Width scaling and dynamical regimes.** (a) $\alpha - 1 \sim m^{1.18}$ for MSE, with increasing curvature at large widths. (b) The transition width m^* depends on absolute overparameterization, not m/d . (c) At EoS, the per-neuron activation switch rate is width-independent, confirming extensive $O(m)$ total mode coupling.

1. **Sub-EoS** ($\lambda_{\max} < 2/\eta$): Per-neuron activation switch rate $\sim m^{-0.5}$, total mode coupling $O(\sqrt{m})$, and the spectral crossover formula applies with perturbative corrections.
2. **At EoS** ($\lambda_{\max} \approx 2/\eta$): Per-neuron switch rate is width-independent, total mode coupling is $O(m)$, and the simple power-law drift model $\sim \eta^\alpha$ develops significant curvature in log-log space.

Width scaling of α . For MSE loss, the drift exponent grows with width as $\alpha - 1 \sim c \cdot m^{1.18}$ (E19). The power-law quality degrades systematically from $R^2 = 0.999$ at width 16 to $R^2 = 0.887$ at width 192, consistent with the transition from perturbative to non-perturbative dynamics.

Width-dimension transition. The transition between regimes depends on the *absolute overparameterization* rather than a fixed m/d ratio (E22). For input dimensions $d \in \{10, 20, 40\}$, the transition width satisfies $m^*/d = 6.0, 3.0, 1.0$ respectively—the transition occurs earlier (at smaller m/d) for larger d because even modest widths provide sufficient parameters relative to the training data constraints.

6 Discussion

Our results provide a unified spectral theory for why gradient descent navigates non-convex neural network landscapes. The key insight is that conservation laws from the network’s symmetry group serve as “guide rails” during early training, confining trajectories to structured submanifolds. At the Edge of Stability, discrete gradient descent breaks these laws in a structured way—the spectral crossover formula (Theorem 4) explains the precise power-law scaling of the drift from first principles.

Cross-entropy as a self-regularizing loss. The spectral compression mechanism (Theorem 6) reveals that CE loss has a *built-in* regularization property: softmax probability concentration exponentially shrinks the Hessian spectrum during training, preventing the extensive mode coupling that drives α upward in the MSE case. The compression timescale $\tau = \Theta(1/\eta)$ is n -independent in the overparameterized regime, connecting to the NTK theory in a novel way.

Practical implications. Our theory suggests that learning rate schedules should respect the EoS boundary: operating at $\eta \approx 2/\lambda_{\max}$ maximizes structured conservation law breaking and improves training. For CE loss, the self-regularization implies robustness to learning rate choice, consistent with empirical observations.

Open problems. Several directions remain: (i) computing c_k at EoS with extensive mode coupling, where the independent-mode decomposition breaks down; (ii) extending the theory beyond 2-layer networks, where the mean-field quasi-convexity result (Theorem 1) applies but the spectral analysis requires multi-layer generalizations; (iii) connecting conservation law breaking directly to generalization bounds; (iv) bridging to the percolation and tropical Morse perspectives on mode connectivity.

Acknowledgments. Computational experiments were performed on a consumer-grade CPU (Intel i5-1038NG7, 16GB RAM) using PyTorch 2.2.2, demonstrating that rigorous ML theory research is accessible without GPU resources.

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A Full Proofs

A.1 Proof of Theorem 1 (Conservation Laws)

Consider an L -layer ReLU network $f(x; \theta) = W_L \sigma(W_{L-1} \sigma(\cdots \sigma(W_1 x)))$ with no bias terms. ReLU is positively 1-homogeneous: $\sigma(\alpha z) = \alpha \sigma(z)$ for $\alpha > 0$.

Step 1: Rescaling invariance. For any $l \in \{1, \dots, L-1\}$ and $\alpha > 0$, the transformation $W_l \rightarrow \alpha W_l$, $W_{l+1} \rightarrow \alpha^{-1} W_{l+1}$ preserves the network function:

$$W_{l+1} \sigma(W_l x) = (\alpha^{-1} W_{l+1}) \sigma((\alpha W_l) x) = \alpha^{-1} W_{l+1} \cdot \alpha \cdot \sigma(W_l x) = W_{l+1} \sigma(W_l x), \quad (9)$$

using the 1-homogeneity of σ . Since f is invariant, $\mathcal{L}$ is also invariant.

Step 2: Infinitesimal symmetry. Differentiating the invariance $\mathcal{L}(\dots, \alpha W_l, \alpha^{-1} W_{l+1}, \dots) = \mathcal{L}(\dots, W_l, W_{l+1}, \dots)$ with respect to α at $\alpha = 1$:

$$\text{tr} \left(W_l^\top \frac{\partial \mathcal{L}}{\partial W_l} \right) = \text{tr} \left(W_{l+1}^\top \frac{\partial \mathcal{L}}{\partial W_{l+1}} \right). \quad (10)$$

Step 3: Conservation. Under gradient flow, $\frac{d}{dt} \|W_l\|_F^2 = 2 \text{tr}(W_l^\top \dot{W}_l) = -2 \text{tr}(W_l^\top \frac{\partial \mathcal{L}}{\partial W_l})$. By (10), this rate is identical for all l . Therefore:

$$\frac{d}{dt} C_l = \frac{d}{dt} (\|W_{l+1}\|_F^2 - \|W_l\|_F^2) = 0. \quad \square \quad (11)$$

A.2 Proof of Theorem 2' (Mean-Field Quasi-Convexity)

Theorem (Mean-Field Quasi-Convexity on M_C). For a 2-layer ReLU network without bias, with MSE loss on n data points in $\mathbb{R}^d$, in the mean-field limit ($m \rightarrow \infty$): every local minimum of $\mathcal{L}$ restricted to M_C is a global minimum.

Proof. Following Chizat and Bach [2] and Mei et al. [10], represent the infinite-width network as a measure ρ on $\Omega = \mathbb{R}^d \times \mathbb{R}^K$:

$$f_\rho(x) = \int_{\Omega} a \cdot \sigma(w^\top x) d\rho(w, a). \quad (12)$$

Step 1: The MSE risk $R(\rho) = \frac{1}{2n} \sum_i \|f_\rho(x_i) - y_i\|^2$ is *convex* in ρ , since f_ρ is linear in ρ and $\|\cdot\|^2$ is convex.

Step 2: The conservation constraint $C(\rho) = \int_{\Omega} (\|a\|^2 - \|w\|^2) d\rho = c$ is a linear functional of ρ , so the constraint set $M_C^\infty = \{\rho : C(\rho) = c\}$ is an affine subspace—in particular, convex.

Step 3: A convex function restricted to a convex set has no spurious local minima. $\square$

Remark 9. The remaining gap is finite-width convergence: showing that the discrete measure ρ_m on M_C converges to the global minimizer of R on M_C^∞ at rate $O(1/\sqrt{m})$. Standard propagation-of-chaos results apply since the constraint is linear.

A.3 Proof of Theorem 2 (Drift Decomposition)

Proof. Under gradient descent: $W_l(t+1) = W_l(t) - \eta \frac{\partial \mathcal{L}}{\partial W_l}(t)$. Expanding:

$$\|W_l(t+1)\|_F^2 = \|W_l(t)\|_F^2 - 2\eta \operatorname{tr}\left(W_l^\top \frac{\partial \mathcal{L}}{\partial W_l}\right) + \eta^2 \left\| \frac{\partial \mathcal{L}}{\partial W_l} \right\|_F^2. \quad (13)$$

Taking the difference between layers $l+1$ and l :

$$\begin{aligned} C_l(t+1) &= C_l(t) - 2\eta \underbrace{\left[\operatorname{tr}\left(W_{l+1}^\top \frac{\partial \mathcal{L}}{\partial W_{l+1}}\right) - \operatorname{tr}\left(W_l^\top \frac{\partial \mathcal{L}}{\partial W_l}\right) \right]}_{=0 \text{ by (10)}} \\ &\quad + \eta^2 \left[\left\| \frac{\partial \mathcal{L}}{\partial W_{l+1}} \right\|_F^2 - \left\| \frac{\partial \mathcal{L}}{\partial W_l} \right\|_F^2 \right]. \end{aligned} \quad (14)$$

The $O(\eta)$ term vanishes by the same symmetry argument as Theorem 1 (the traces are equal for discrete weights as well, since the identity (10) holds at any θ). The remaining $O(\eta^2)$ term gives the exact per-step drift. Summing over T steps yields the gradient imbalance sum $S(\eta)$. $\square$

A.4 Proof of Theorem 3 (Linear Network Same α)

For a 2-layer linear network $f(x) = W_2 W_1 x$, the MSE loss gradient with respect to each layer is:

$$\frac{\partial \mathcal{L}}{\partial W_1} = -\frac{1}{n} W_2^\top (Y - W_2 W_1 X) X^\top, \quad (15)$$

$$\frac{\partial \mathcal{L}}{\partial W_2} = -\frac{1}{n} (Y - W_2 W_1 X) (W_1 X)^\top. \quad (16)$$

Decomposing in the data covariance eigenbasis $\Sigma_x = U_x \Lambda_x U_x^\top$ yields independent 1D problems. For each mode k with effective error $e_k(t)$ and Hessian eigenvalue $\lambda_k = 2\lambda_{x,k}\sigma_{k,0}^2$, the error evolves as:

$$e_k(t) = e_k(0) \cdot (1 - \eta\lambda_k)^t = e_k(0) \cdot \rho_k^t. \quad (17)$$

The gradient imbalance for mode k contributes $\sum_t e_k(t)^2 = e_k(0)^2 (1 - \rho_k^{2T}) / (1 - \rho_k^2)$, which is exactly the spectral crossover formula with $c_k \propto e_k(0)^2 \lambda_{x,k}^2$. The resulting drift exponent $\alpha = 1.10$ matches the ReLU case because both share the same Hessian spectral structure (ReLU adds mode coupling but does not change the leading-order spectral decomposition).

A.5 Proof of Theorem 4 (Spectral Crossover Formula)

Proof. From Theorem 2, $S(\eta) = \sum_t \delta(t)$ where $\delta(t) = \|\partial \mathcal{L} / \partial W_{l+1}\|_F^2 - \|\partial \mathcal{L} / \partial W_l\|_F^2$.

For linear networks, the mode decomposition (Appendix A.4) gives $\delta(t) = \sum_k c_k \cdot e_k(t)^2 / e_k(0)^2$ where the constant c_k captures the mode-dependent gradient imbalance structure.

Summing over time:

$$S(\eta) = \sum_k c_k \sum_{t=0}^{T-1} \frac{e_k(t)^2}{e_k(0)^2} = \sum_k c_k \sum_{t=0}^{T-1} \rho_k^{2t} = \sum_k c_k \cdot \frac{1 - \rho_k^{2T}}{1 - \rho_k^2}. \quad (18)$$

Since $1 - \rho_k^2 = 1 - (1 - \eta\lambda_k)^2 = \eta\lambda_k(2 - \eta\lambda_k)$:

$$S(\eta) = \sum_k c_k \cdot \frac{1 - (1 - \eta\lambda_k)^{2T}}{\eta\lambda_k(2 - \eta\lambda_k)}. \quad (19)$$

For ReLU networks, the activation pattern couples modes, but for sub-EoS learning rates where the switch rate is $O(m^{-0.5})$, the independent-mode decomposition holds to first order with perturbative corrections. The formula is exact for linear networks and approximate (within 14–27%) for ReLU. $\square$

A.6 Proof of Theorem 6 (Spectral Compression)

Proof. The CE Gauss-Newton Hessian is $H_{\text{CE}}(t) = \frac{1}{n} J(t)^\top S(p(t)) J(t)$ where $S = \text{block_diag}(S_1, \dots, S_n)$ with $S_i = \text{diag}(p_i) - p_i p_i^\top$.

Each block S_i is PSD with $\lambda_{\max}(S_i) = \max_k p_{i,k}(1 - p_{i,k})$. By the variational characterization:

$$\lambda_{\max}(H_{\text{CE}}) = \max_{\|v\|=1} \frac{1}{n} \sum_i (J_i v)^\top S_i (J_i v) \quad (20)$$

$$\leq \max_{\|v\|=1} \frac{1}{n} \sum_i \lambda_{\max}(S_i) \|J_i v\|^2 \quad (21)$$

$$\leq \max_i \lambda_{\max}(S_i) \cdot \lambda_{\max}\left(\frac{1}{n} J^\top J\right). \quad (22)$$

When the correct class dominates ($q_i > 1/2$), $\max_k p_{i,k}(1 - p_{i,k}) = q_i(1 - q_i)$, yielding the bound. Under gradient descent on CE, $q_i(t)$ satisfies the logistic ODE $dq_i/dt = q_i(1 - q_i)g_i(t)$ with $g_i > 0$, ensuring $q_i \rightarrow 1$ and hence $q_i(1 - q_i) \rightarrow 0$ exponentially. $\square$

A.7 Proof of Theorem 5 (Mode Coefficients)

Proof. For a 2-layer linear network with data covariance eigendecomposition $\Sigma_x = U_x \Lambda_x U_x^\top$, the problem decomposes into independent modes. In mode k , the effective parameterization is $\sigma_k = \sigma_{2,k} \sigma_{1,k}$ with gradient norms:

$$\left| \frac{\partial \mathcal{L}}{\partial \sigma_{1,k}} \right|^2 = (\sigma_k - \sigma_k^*)^2 \sigma_{2,k}^2 \lambda_{x,k}^2, \quad (23)$$

$$\left| \frac{\partial \mathcal{L}}{\partial \sigma_{2,k}} \right|^2 = (\sigma_k - \sigma_k^*)^2 \sigma_{1,k}^2 \lambda_{x,k}^2. \quad (24)$$

The gradient imbalance for mode k is $\delta_k = e_k^2 \lambda_{x,k}^2 (\sigma_{1,k}^2 - \sigma_{2,k}^2)$, where $e_k = \sigma_k - \sigma_k^*$. The weight imbalance $(\sigma_{1,k}^2 - \sigma_{2,k}^2)$ is the conservation quantity for this mode, which evolves only at $O(\eta^2)$ due to Theorem 2.

At leading order, the time-summed contribution is dominated by the initial error:

$$c_k = \sum_t \delta_k(t) \propto e_k(0)^2 \lambda_{x,k}^2 \cdot (\sigma_{1,k}(0)^2 - \sigma_{2,k}(0)^2). \quad (25)$$

For Kaiming initialization with balanced layers ($\sigma_{1,k} \approx \sigma_{2,k}$), the imbalance develops from $O(\eta^2)$ discretization error and is proportional to $e_k(0)^2 \lambda_{x,k}^2$, independent of the initialization scale $\sigma_{k,0}$. $\square$

A.8 Proof of Proposition 7 (Compression Timescale)

Proof. Under gradient descent on CE, the correct-class probability for sample i evolves as:

$$\frac{dq_i}{dt} = q_i(1 - q_i) \cdot g_i(t), \quad g_i(t) = \frac{\eta}{n} \sum_j \kappa(x_i, x_j) \cdot r_j(t), \quad (26)$$

where $\kappa(x_i, x_j) = J(x_i)^\top J(x_j)$ is the NTK kernel and $r_j = 1 - q_j$ are residuals.

In the overparameterized regime ($m \gg n \log n / \lambda_0$), the NTK matrix K with $K_{ij} = \kappa(x_i, x_j)$ satisfies $\lambda_{\min}(K) = \Theta(1)$ [5, 7]. Its trace is $\text{tr}(K) = \sum_i \kappa(x_i, x_i) = n \cdot C_{\text{arch}}$ where $C_{\text{arch}} = O(1)$.

The mean convergence rate involves contributions from same-class samples via cross-kernel entries. For sample i with class y_i , the aggregate growth rate from $\sim n/K$ same-class samples gives:

$$g_i \approx \eta \cdot C_{\text{cross}} / K + O(\eta/n), \quad (27)$$

where $C_{\text{cross}} = \mathbb{E}_{x, x'}[\kappa(x, x')]$ depends on architecture and data distribution but not n . The $O(\eta/n)$ self-term becomes negligible for large n .

The compression timescale is $\tau = 1/g_{\min} \approx K/(\eta \cdot C_{\text{cross}}) = \Theta(1/\eta)$, independent of n . $\square$

B Extended Experimental Results

All experiments use 2-layer networks (unless noted), Gaussian mixture data ($n = 200$, $d = 20$, $K = 5$, separation 2.0), seeds $\{42, 137, 256, 512, 1024\}$, and full-batch gradient descent. Results are averaged over seeds with standard errors reported.

Table 1: Summary of all 23 key experiments. Full configurations and per-seed results available in the code repository.

#	Name	Key Result	Theory Link	Session
E1	Conservation verification	Drift $< 0.003\%$	Thm 1	1
E2	Conservation with bias	Bias breaks conservation	Thm 1	1
E3	Drift vs. learning rate	Drift $\sim \eta$ scaling	Thm 2	1
E4	EoS conservation breaking	5500 $\times$ drift increase	Thm 2	2
E5	Drift scaling law	$\alpha = 1.16$, $R^2 > 0.99$	Thm 4	2
E6	Depth dependence	α : 1.07 (2L) to 1.72 (8L)	Thm 4	3
E7	Optimizer dependence	Adam: $\alpha = 0.56$	Thm 4	3
E8	Spectral universality	14–27% prediction error	Thm 4	5
E9	Linear-ReLU gap	2.2% switch rate difference	Thms 3,4	5
E10	Activation coupling	Smooth α transition	Thm 4	5
E11	Interpolated activation	α varies with homogeneity	Thm 4	5
E12	Loss function interaction	Non-additive 3-factor decomp.	Thms 4,6	6
E13	CE clamping mechanism	CE $\alpha \approx 1.0$ at all widths	Thm 6	6
E14	Interaction with width	CE regularization grows with m	Thm 6	6
E15	Width switch rate	Per-neuron rate m -independent at EoS	Thm 8	7
E16	Time-dependent Hessian	CE $R = 0.988$ at $t = 250$	Thm 6	7
E17	CE clamping effect	CE clamps $\alpha \approx 1.0$	Thm 6	7
E18	CE Hessian evolution	24 $\times$ compression, n -indep.	Thm 6	8
E19	MSE fine width sweep	$\alpha - 1 \sim m^{1.18}$	Thm 8	8
E20	Linear c_k validation	$R = 0.847$	Thm 5	8
E21	ReLU c_k validation	$R > 0.80$ at all η	Thm 5	9
E22	Width-dimension transition	m^*/d varies: 6.0, 3.0, 1.0	Thm 8	9
E23	τ vs. learning rate	$\tau = 1.33/\eta + 29$, $R^2 = 0.988$	Prop. 7	9

C Reproducibility

Hardware. Intel Core i5-1038NG7 (4 cores, 2.0 GHz), 16 GB RAM, CPU only (no GPU).

Software. Python 3.12.7, PyTorch 2.2.2, NumPy 1.26.4, Matplotlib 3.9.2.

Random seeds. All experiments use seeds $\{42, 137, 256, 512, 1024\}$ (or a subset of 3 seeds for computationally intensive experiments). Seeds are set for both Python’s random module and PyTorch.

Code availability. All experiment scripts, the shared utility library, and configuration files are available at <https://github.com/danielxmed/TheLocalMinimumParadox>. Each experiment saves a `config.json` file with the complete configuration and a `results.json` file with processed results, enabling exact reproduction.

Computational cost. Individual experiments run in 30 seconds to 15 minutes on the hardware above. The full suite of 23 experiments requires approximately 4 hours of total CPU time.